

SG GoAssist Autonomous Passenger Assistance System

Draft project report

Singapore BusTech Grand Challenge 2026

IHL Student Category

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Institution National University of Singapore

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Draft for group completion. Incomplete results and budget fields remain blank.

Abstract

SG GoAssist aims to help passengers request and receive boarding or alighting assistance on an autonomous bus. Our prototype combines an accessible request interface, local perception, a deterministic assistance controller and a model ramp. The intended outcome is a complete journey in which the system acknowledges a request, checks safety conditions, provides assistance and verifies that the ramp is stowed before departure. [1, 2]

The current project has software and firmware for this assistance loop, a detailed bus and ramp CAD model, and an ESP32-S3 bench setup that has produced valid distance readings. The software includes passenger and operator interfaces, assistance-case management, telemetry, interlocks and audit logging. CAD records document a discrete motion-clearance study. These are separate forms of progress and do not yet establish a completed physical system. [3–7]

The next priority is physical integration and controlled validation. Ramp actuation, obstacle coverage, model accuracy, repeatability and user experience still require measured results. The results and feedback sections below remain open until those activities are complete. The report follows the four-week prototype plan and its D1 to D9 deliverables. [1]

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1 Introduction

Problem and intended users

The project addresses the assistance tasks that remain around an autonomous bus journey: identifying the correct service, requesting enough boarding time, accessing a ramp and understanding whether help is on its way. A passenger standing near a bus stop may be waiting for another service, so presence alone cannot establish boarding intent. Visible mobility aids also cannot represent every assistance need. [1, 2]

The proposed system supports wheelchair and personal mobility aid users, passengers using walking aids, older passengers, people with temporary or non-visible needs, and passengers who benefit from audio, visual or haptic information. These are intended user groups. Representative-user validation remains an unfinished project activity. [1, 2]

Project objectives

Objective	Planned evidence
Recognise observable assistance cues	Held-out class metrics and examples of errors for the perception models.
Act on explicit passenger intent	Correct request acknowledgement, bus assignment and cancellation in controlled scenarios.
Provide a verified mechanical response	Safe model-ramp deployment, boarding hold and confirmed retraction.
Measure accessibility outcomes	Comparison with a manual-assistance baseline and consented participant feedback.

Minimum viable prototype

The action plan defines one repeatable journey using a stop unit or lightweight app, a mock bus, a boarding fixture, safety sensors and an operator dashboard. A second scenario requests additional time and audio or visual guidance without a ramp. This demonstrates that an explicit request can support needs that a camera cannot infer. [1]

The prototype scope excludes a road-capable autonomous vehicle, a full-scale certified ramp and claims of reliable perception in every public environment. Production transport feeds, city-wide infrastructure and advanced walking guidance are outside the core demonstration. The repository contains additional journey-planning and assistant features, but these do not replace the core integration and validation tasks. [1, 4]

Development of the concept

The initial expression of interest proposed RFID requests, camera/LiDAR recognition and pressure or magnetic sensing. The later design gives a tactile stop button a central role, while the action plan

retains an app and optional NFC. The implemented software accepts several input types under one assistance-case contract. For the first physical demonstration, the team still needs to freeze the actual input devices and sensor configuration. [1–3, 9]

2 Design alternatives and original contributions

Alternatives and selection rationale

The design progressed from the expression of interest to the later request-driven prototype. The alternatives below explain the current direction without treating earlier concepts as completed hardware. [1, 2, 9]

Alternative	Tradeoff	Selected approach
RFID or NFC only	Simple explicit request, but depends on obtaining and carrying a card.	Optional input alongside a tactile button and app.
Camera-triggered ramp	Recognises observable cues, but cannot establish which bus a passenger intends to board.	Use perception for cues and dwell; require confirmed ramp intent.
Pressure or magnetic presence sensing	Adds redundancy but can confuse passengers, luggage and passers-by.	Treat presence as supporting evidence, with geometric safety checks.
App-only request	Flexible selection and feedback, but excludes passengers without a suitable device.	Keep a stop-based tactile request route.
Full-scale vehicle actuation	Closer to deployment, with major mechanical and safety demands.	Demonstrate on a low-voltage mock bus with measured interlocks.

Team contribution and use of existing technology

The team contribution is the integration of explicit passenger intent with an assistance-case workflow, deterministic safety checks, local actuator verification and accessible feedback. The repository implements the request and verification logic, and the CAD work develops the centre-door boarding mechanism concept. These are the elements to demonstrate and explain as project work. [2, 3, 6]

ESP32 hardware, ToF sensors, RFID/NFC, ArUco pose estimation and pretrained model architectures are existing technologies. The bus model is reference-inspired, and proposed mechanisms are not OEM-verified. The team should credit these foundations and identify any adapted code, models or designs in the final references. No claim of worldwide novelty or patentability is made. [6, 8, 9]

The official rubric allocates 20% to design and methodology, 30% to execution, 30% to originality and creativity, and 20% to project presentation. Only original elements of an improvement on an existing design receive originality credit. Final contribution records and comparisons should therefore identify precisely what the group designed and implemented. [11, p. 16]

3 Methods and system design

System architecture

Layer	Role in the prototype
Passenger and stop unit	App, tactile button or optional NFC records the requested assistance. Status returns through accessible feedback.
Local perception	An edge computer interprets camera observations. Distance and obstruction sensors support geometric checks.
Assistance controller	The backend retains intent, assigns the bus and stop, checks capability and manages the assistance case.
Mock bus controller	ESP32 firmware receives commands, repeats local safety checks and reports actuator and limit-switch status.
Operator and evidence	The console supports confirmation and fault recovery. Metrics and audit events record the outcome.

Intended demonstration sequence

A passenger requests ramp assistance and, where needed, extra time. The system acknowledges the request for the selected bus. Perception can support positioning or dwell decisions, while the request remains authoritative. The model bus then docks at the marked boarding fixture and confirms that it is stopped securely. [1, 3]

The ramp area is visually marked. Before deployment, the controller checks the bus and stop match, available capability and space, fresh telemetry, stopped and brake signals, open door, clear path and controller health. The local controller repeats relevant checks while the ramp moves. “Ready” requires both completed actuation and a deployed limit-switch indication. [3, 5]

The system maintains assistance during boarding. A completion or cancellation event initiates a safe closure sequence. A case reaches its terminal state only after stowed-ramp verification. Departure also requires a stowed ramp and clear departure conditions. Faults or ambiguous states remain visible to the operator. [3, 5]

Safety and passenger control

Sensor-only detections cannot confirm a ramp request. They enter a confirmation path, while reversible extra dwell may proceed from a detection. Missing or stale safety data blocks movement. The prototype design also requires a physical emergency stop and independent electrical interlocks before physical demonstrations. Their installation and behavior must be checked on the assembled model. [2, 3, 5]

Privacy boundary

The reference edge observer processes frames locally and sends class/confidence metadata. It does not save or transmit camera frames. The system is intended to recognise objects and assistance cues without identifying people or diagnosing disabilities. Any future remote-video function from the earlier concept requires a separate privacy and consent design. [2, 3, 8]

4 Software implementation results

Implemented software scope

The repository provides a passenger application, a backend and shared message contracts. The assistance controller connects explicit requests, observations, vehicle capabilities, safety telemetry and actuator status. It preserves separate passenger intents while using idempotent commands to avoid duplicate actuator actions. [3, 4]

Area	Current implementation	Remaining evidence
Passenger app	Requests, status, cancellation, nearby stops, journeys and accessibility options.	Physical-device and representative-user checks.
Operator console	Case queue, safety state, confirmation, escalation and retry.	Operator recovery trials with the connected model.
Assistance backend	Case states, safety gates, command status, persistence and audit events.	Timestamped integrated hardware runs.
Mock autonomy	Route-state simulation and marker/ToF docking integration points.	Measured physical docking performance.
Automated checks	Repository tests cover request handling, interlocks, intent, persistence and integration paths.	Attach a dated release-test run and its build revision.

Evidence boundaries

The readiness matrix records automated software checks and separately labels competition fixtures. Demonstration arrivals, selected stop amenities, shelter information and advisories must retain their fixture labels. They are not evidence of a live transport-provider integration. [4]

The multilingual assistant includes deterministic handling and an optional on-device Qwen path. Connected ARM64-device verification and physical speech-input testing remain release gates. The assistant can interpret requests and explain information, but safety controllers retain authority over ramp, door and vehicle actions. [4]

Software baseline for the demonstration

The final release should identify a fixed source revision and configuration, reproduce the nominal and fault scenarios, and archive the console/event output. Software checks support implementation confidence, but physical trials must establish the behavior of the assembled prototype.

Release revision and dated test summary

To complete

5 Mechanical and electronics results

Mechanical model

Robobus v2 includes a centre-door access concept, plug-and-slide door leaves, a translating and tilting ramp, a bridge and a kerb fixture. Native Fusion and STEP exports exist for travel and deployed states. The CAD validation record reports 189 valid solid bodies and eight native as-built joints. [6]

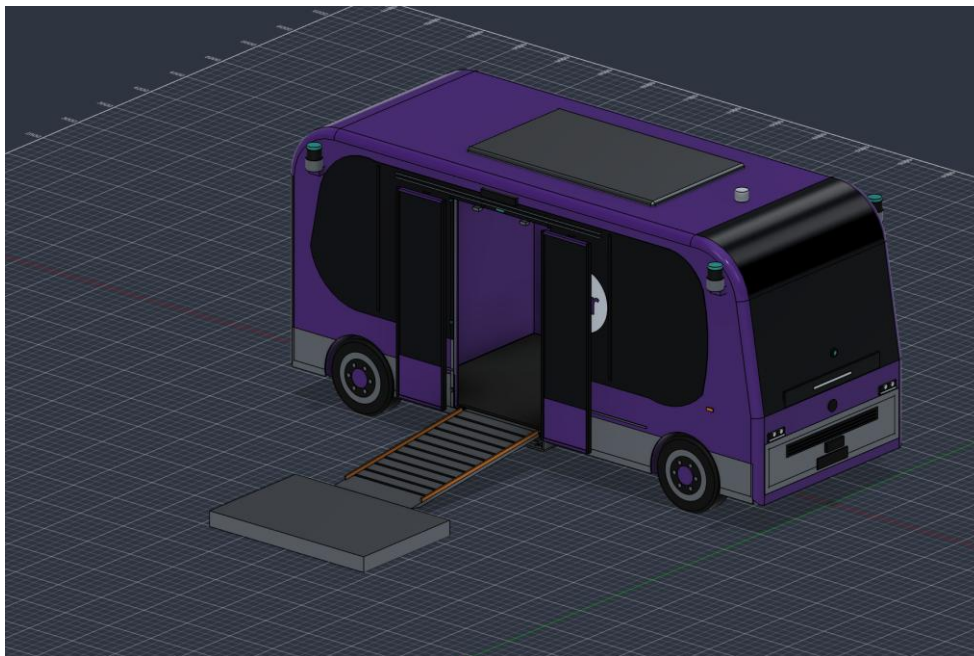


Figure 1 CAD boarding-ready concept. Proposed geometry, not a photograph of the assembled prototype. [6]

The recorded clearance study sampled 73 poses with zero moving-part intersection volumes reported. A separate passage check addressed a static doorway obstruction. This is a discrete CAD packaging result. It does not verify continuous collision clearance, structural strength, load capacity or physical reliability. [6]

Electronics bench progress

On 23 September, the ESP32-S3 test sketch compiled and uploaded. The connected sensor was identified as a VL53L0X. After initial invalid readings and a full power cycle, the recorded run

produced 27 consecutive valid samples from 7639 to 15095 ms, spanning 30 to 136 mm. This establishes live ranging only. No known-distance accuracy result or automatic ramp-stop result follows from that test. [7]

The current sensor returns one distance, while the integrated mock-bus firmware describes VL53L5CX 8 × 8 sensing. These configurations are not interchangeable. The team must reconcile firmware, mounting and coverage before claiming that the whole ramp area is protected. Laser wiring/setup drawings exist, but the ToF test keeps laser outputs off and does not validate projection or switching. [5, 7, 10]

6 Machine learning methods and status

Planned model functions

Module	Bounded decision	Progress and next evidence
ML 1 assistance cues	Detect observable mobility aids and relevant waiting-zone cues.	Training/evaluation scripts and label guidance exist. Complete dataset, exported weights and held-out results.
ML 2 ramp zone	Classify scene objects and combine them with geometric occupancy.	Obstacle-fusion software and sensor rules exist. Validate semantics and physical coverage against the obstruction matrix.
ML 3 boarding state	Estimate active, complete or uncertain boarding from temporal features.	The action plan defines the model and a fixed-dwell fallback. Complete sequence labels, training and evaluation.
Docking	Estimate pose relative to a marker and verify range.	ArUco/ToF observer code exists. Calibrate and measure repeated physical docking.

The planned models support limited decisions. A cue detector cannot diagnose a disability or authorise ramp movement. ML 2 must operate within the deterministic safety boundary, and ML 3 must retain a sensor veto and a fallback dwell rule. ArUco docking is a marker-based vision method rather than a trained ML model. [1, 5, 8]

Dataset and evaluation method

Use consented or appropriately licensed imagery, document class definitions and split recordings by session or location. Keep the test split separate from training and threshold selection. Record the dataset version, model checksum, test conditions and device used for timing. The ML 1 package includes tools to record provenance and compute precision, recall, mAP and inference speed. [1, 8]

The action plan proposed 300–500 ML 1 images, while the later repository guide proposes 800–1,200. These are planning targets, not dataset counts. The final dataset scope and achieved coverage should be documented below. [1, 8]

Model record	ML 1	ML 2	ML 3
Dataset version and split			
Selected model and weights			
Test sample count			
Precision and recall or state accuracy			
Latency or inference speed			
Known failure cases			

Blank cells are reserved for measured model results and completed model records.

7 Deliverable readiness

The following assessment follows D1 to D9 in the prototype action plan. “Implemented” describes an available software/design artifact. It does not imply physical integration or completion of acceptance trials. [1, 3–8]

ID and deliverable	Current position	Completion requirement
D1 Architecture and interfaces	Documented and implemented in shared contracts and controller logic.	Freeze hardware-specific interfaces and the demonstration configuration.
D2 Integrated physical prototype	Firmware and CAD exist; ToF bench ranging verified.	Assemble bus, stop unit and fixture; record 10 cold-start journeys.
D3 Passenger app and dashboard	Implemented, with automated checks recorded in project documentation.	Capture the connected request, status, cancellation and escalation journey.
D4 ML 1 package	Training/evaluation pipeline and labels prepared.	Dataset manifest, trained weights, metrics, latency and model card.
D5 ML 2 and safety fusion	Software interfaces and safety-fusion logic prepared.	Semantic results plus physical obstruction and uncertainty trials.
D6 ML 3 or fallback	Model and fallback defined in the plan.	Validate the selected dwell method and record premature-clear outcomes.
D7 Mechanical and electrical	CAD/export package, wiring drawings and sensor bring-up available.	As-built wiring, BOM, power budget, actuator/interlock and cycle tests.

ID and deliverable	Current position	Completion requirement
D8 Validation and accessibility	Scenario matrix and metric collection paths defined.	Measured baseline comparison, limitations and consented feedback.
D9 Submission package	Draft report and presentation prepared.	Final evidence, compliant media, source/model archive and rehearsal.

Scope decisions to close

Confirm the final sensor and actuator arrangement, the role of projected lines, and whether automatic driving remains in the live demonstration. The action plan permits scope reductions, including NFC and automatic driving, while preserving the request path, ramp-zone assessment, interlocks, physical ramp and complete evidence trail. [1]

The earlier proposal and later firmware also differ in their treatment of light debris and motion resumption. Until the selected rule is documented and physically validated, an occupied or uncertain ramp zone should remain a blocking demonstration condition. [1, 2, 5]

8 Validation methods and pending results

Controlled test protocol

Freeze the build, wiring and calibration before testing. Record each scenario, input source, expected response, actual response, timestamp and any intervention. Test nominal boarding and alighting, cancellation, duplicate requests, obstacles, ambiguous perception, sensor failure, network loss and non-ramp assistance. Preserve the audit log and metrics snapshot for each run. [1, 3, 5]

For the baseline, repeat the same scripted journey with a person handling the assistance request, then with SG GoAssist. Compare acknowledgement time, successful completion, manual interventions, unnecessary waiting and perceived confidence. Participant counts, task conditions and failures must accompany the comparison. [1]

Acceptance targets and measured outcomes

Measure	Proposed acceptance target	Measured result
Request delivery	20/20 local trials in the action plan; later traceability calls for 30/30 per explicit source.	
Response time	Within 2 s in the action plan; later traceability specifies $P95 < 2$ s.	
ML 1	At least 90% controlled wheelchair/PMA recall and at least 10 FPS.	
Obstacle and interlock tests	Zero missed defined obstructions; zero unsafe movements in at least 50 interlock/obstruction trials.	

Measure	Proposed acceptance target	Measured result
Actuation	50 ramp/door cycles without binding or interlock bypass.	
Docking	10 consecutive trials within demonstrated ramp reach.	
Boarding/dwell	No premature CLEAR in defined sequence tests; fallback available.	
Complete journey	At least 9 of 10 nominal cold-start journeys without manual intervention.	

Targets are drawn from the action plan and later traceability document. Blank results are intentional. Reconcile the request-trial count and latency definition in the final test protocol. [1, 5]

Measured baseline comparison

To complete

Integrated demonstration results and failure analysis

To complete

9 Discussion and remaining work

Analysis of current results

The CAD study supports the feasibility of the proposed packaging at sampled positions, and the sensor log shows successful communication and ranging after recovery. Neither result demonstrates a safe complete journey. The sensor-type mismatch, untested actuation and incomplete model evidence prevent a reliable estimate of end-to-end performance. Measured user benefit remains an open research outcome. [6, 7]

Immediate integration work

First, freeze the demonstration hardware and align firmware with the actual distance sensor. Calibrate known distances and the monitored ramp area. Connect and verify the chosen ramp actuator, its driver and power supply, position feedback, request input and independent stop/interlock functions. Then connect the physical telemetry and actuator results to the existing assistance case. [1, 5, 7]

Next, reproduce one nominal journey before expanding the scenario matrix. Validate the selected perception models and dwell method on held-out data and the actual demonstration device. Use the fixed extended-dwell fallback if ML 3 cannot meet its evidence requirements. The final stage is repeatability testing, evidence capture and rehearsal. [1]

Risk	Effect on the demonstration	Planned response
Sensor coverage mismatch	One valid range could be mistaken for a clear ramp area.	Confirm coverage and use a documented conservative blocking rule.
Actuator or mechanism fault	Ramp fails to reach or leave its intended position.	Verify limits, power, mechanical clearance and safe operator recovery.
Insufficient ML evidence	Recognition claims cannot be substantiated.	Reduce classes if needed and report held-out results and failure cases.
Late integration	Subsystem progress does not produce a complete journey.	Freeze interfaces, integrate the core path and defer peripheral features.
Overstated impact	Scenario success is mistaken for user-validated benefit.	Separate bench, software, scenario and representative-user evidence.

Remaining schedule

Planned milestone	Work to close	Actual completion
25–26 September	Integration, priority fault checks and evidence capture.	
27 September	Validation and content freeze from the original plan.	
28–29 September	Report evidence audit, poster artwork and submission review.	
30 September	Project report and display poster artwork due.	
7 October	25-minute pre-event prototype showcase and demonstration.	
15 October	Main-event slideshow and video submission due.	
24 October	Main event: 5-minute talk, up to 5-minute video and 5-minute Q&A.	

25–29 September entries are internal planning. Official deadlines and event timing follow the 18 August update. Event dates/venues remain tentative. The 24 October presentation applies if selected for the main event. [1, 11, pp. 14–18]

10 Proposed budget and team records

The following sections are reserved for the group to complete after decisions and trials. Empty fields should remain empty until the associated record or result is available.

Final team responsibilities

Role from the action plan	Member	Final contribution
Systems and software lead		
ML and computer vision lead		
Stop unit and sensing lead		
Bus electronics and control lead		
Mechanical and validation lead		

Proposed prototype budget

Item or category	Quantity	Unit cost SGD	Proposed total SGD
Controllers and communications			
Camera and distance sensors			
Ramp actuator and driver			
Switches and feedback devices			
Power and wiring			
Chassis and boarding fixture			
Contingency			
Total proposed budget			

Complete quantities, quotations and totals before submission. Seed funding amount, in-kind items and actual expenditure should be recorded separately.

User and stakeholder feedback

To complete

Final prototype photographs and demonstration link

To complete

Conclusion

SG GoAssist has established the software structure, mechanical concept and initial ranging setup needed to progress toward an integrated assistance demonstrator. The remaining work is to connect

those elements and measure their behavior under nominal, obstructed and failed conditions. Claims about improved independence or passenger satisfaction will depend on the unfinished comparison and feedback work.

Final conclusion after validation

To complete

11 References

Project documents and implementation records supporting this draft are listed below. Bracketed references identify the relevant sources. Repository paths are relative to the GoAssist project root.

1 SG GoAssist Four-Week Prototype Action Plan

SG_GoAssist_Prototype_Action_Plan.pdf. Project window 31 August–30 September 2026. Primary structure, ML workstreams, D1–D9, acceptance targets and schedule.

2 Autonomous Passenger Assistance System Design Document

APAS_Design_Document.docx. Project objective, user needs, proposed physical request and ramp workflow, requirements and fault concepts.

3 Grand Challenge integrated prototype

docs/grand-challenge-integrated-prototype.md. Assistance loop, intent policy, interlocks, persistence, audit and evidence workflow.

4 Feature readiness and project overview

docs/feature-readiness-matrix.md and README.md. Software status, fixtures and physical-device release gates.

5 Requirements and integrated hardware

docs/task-requirements-traceability.md and hardware/INTEGRATED_PROTOTYPE.md. Requirement mapping, mock-bus configuration, safety checks and physical acceptance work.

6 Robobus v2 validation and design exports

cad/VALIDATION_v2.md, cad/README_v2.md, cad/v2_final_report.json and cad/v2_clearance_report.json. Figure 1: cad/Punggol_WeRide_Robobus_v2_deployed.png.

7 ESP32-S3 sensor bring-up and serial evidence

hardware/esp32-s3-tof-test/README.md and artifacts/apas-tof-build/serial-powercycle-test.txt. Verification record dated 23 September 2026.

8 Edge observer and ML 1 package

hardware/edge-observer/README.md, hardware/edge-observer/ml1/README.md, LABEL_GUIDE.md, train_ml1.py and evaluate_ml1.py. Local inference, dataset guidance and reproducible evaluation.

9 Original expression of interest

Singapore BusTech Grand Challenge 2026 - EOI.docx. Initial RFID, camera/LiDAR and pressure/magnetic sensing concept.

10 Circuit and physical setup drawings

artifacts/apas-circuit/APAS_ESP32S3_RYS1230_BC337_Circuit.pdf and output/pdf/APAS_Three_Laser_Physical_Setup.pdf. Proposed wiring and projection setup, separate from the sensor test.

11 Singapore BusTech Grand Challenge 2026 IHL Student Category Updates

Singapore Bus Academy. SGBTGC 2026 - IHL Updates 18 Aug 2026.pdf. 18 August 2026. Pages 8–13: task and success criteria. Pages 14–17: events, rubric and deadlines. Page 18: report content, format and submission requirements.